

Charlotte Tsui

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EDUCATION

The University of North Carolina at Chapel Hill

Expected May 2027

B.S. Computer Science and Data Science

Coursework: Data Structures, Algorithms and Analysis, Operating Systems, Modern Web Programming, Discrete Structures

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C, Swift

Frameworks & Libraries: React, Angular, Next.js, Tailwind, Firebase, Supabase, PostgreSQL, MongoDB

Developer Tools & Platforms: Git, GitHub, Docker, Kubernetes, Vim, Unix/Linux, Figma

PROFESSIONAL EXPERIENCE

Google | *User Experience Engineer Intern*

May 2026 – Aug 2026

- Oversaw end-to-end design and delivery of an **LLM-powered Swift prototyping framework**, cutting the design-to-code iteration cycle from **weeks to hours** for **20+** designers and engineers.
- Built an **AI-assisted prototyping pipeline** that achieved a **90%** compiler success rate on generated views, reducing prototype turnaround time.
- Developed a library of **36+ custom SwiftUI components** to standardize styling and achieve **100%** compliance with Google's Shared Design System across **2 workflows**.
- Co-managed a longitudinal beta test with **6+ UX designers**, instrumenting daily usability logs and triaging issues to improve framework stability.

SAS Data and AI Solutions | *Software Engineering Intern*

May 2025 – May 2026

- Engineered and shipped a large-scale static code analysis tool using **JSCodeShift**, **TypeScript**, and **MongoDB**, analyzing **100K+** lines of code across enterprise React codebases to guide UI library modernization.
- Implemented **automated CI/CD quality gates** (testing, linting, security) with **GitHub Actions**, improving reliability and ensuring compliance across **20+** internal engineering teams.
- Developed **scalable, accessible React components** with comprehensive **Jest** and **Playwright** test suites, supporting **cross-platform** UI consistency.

Centible | *Lead iOS Developer*

Jan 2025 – Present

- Led software development for **Centible**, a personal finance iOS app with **500+ users**, directing a **20-person Agile** engineering team.
- Implemented a premium subscription model via **StoreKit** and **Plaid** linking, supporting a **\$5.5K+/yr** monetization strategy.
- Increased user engagement by **67%** by analyzing **Firebase** user analytics to optimize onboarding and inform feature development.

UNC Department of Computer Science | *Teaching Assistant*

Jan 2024 – Present

- Guided **200+** students through the full **software development lifecycle**, building full-stack applications with **React**, **Next.js**, **TypeScript**, **Drizzle ORM**, and **Supabase**.
- Led instruction on backend development, **API design**, database management, authentication, **AI integration**, and **Git/GitHub** workflows.
- Taught debugging practices and code review standards in **TypeScript** to **improve student code quality**.

PROJECTS

ByteNotes | *TypeScript, Next.js, Supabase, Tailwind*

- Engineered a scalable full-stack CS notetaking platform featuring a markdown editor and an integrated **StackBlitz live compiler** for interactive code execution.
- Implemented user authentication, real-time presence tracking, and event-driven reactions, **optimizing Supabase queries** for low-latency collaboration.
- Built schedule-saving and sharing features with **Firebase** for persistent, multi-device access.

TA Office Hour Scheduler | *Angular, FastAPI, PostgreSQL, OpenAI*

- Developed a full-stack scheduling system to automate TA office hour assignments via **OpenAI API** integration, deployed with **CloudApps** and **CI/CD** pipelines for streamlined updates.
- Built an interactive, grid-based availability interface with real-time feedback and **JSON-based persistence** for data handling.
- Implemented instructor-facing tools and **RESTful API** endpoints supporting dynamic schedule updates and admin controls.